From Missing Energy to Dark Sectors: How the LHC Searches for Invisible Matter

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Abstract

Dark matter is motivated by astrophysical evidence, but the Large Hadron Collider cannot search for it by directly imaging invisible particles inside ATLAS or CMS. It searches through detector-level inference. This review explains how missing transverse momentum, mono-X recoil signatures, mediator models, visible-invisible complementarity, semi-visible jets, emerging jets, and long-lived-particle signatures turn hidden physics into measurable collider observables. The central claim is that LHC dark-matter searches are best understood as disciplined chains linking reconstruction, background control, statistical comparison, and model interpretation. Classic missing-momentum searches remain central, but modern dark-sector searches broaden the program by testing signatures in which hidden particles partly escape, decay visibly, shower in unusual ways, or appear through displaced detector structure. Null results are therefore not empty outcomes: they exclude defined benchmark scenarios and guide the expansion toward signatures that simpler prompt searches may miss.

1. Introduction

Dark matter is one of the clearest signs that the Standard Model is incomplete, but that does not mean every dark-matter experiment asks the same question. Astrophysical and cosmological observations motivate the existence of matter that gravitates without behaving like ordinary luminous matter. Direct-detection experiments ask whether ambient dark matter passing through Earth scatters in a low-background detector. The LHC asks a different question: if dark matter or a related dark sector can be produced in high-energy proton-proton collisions, what visible detector signature would reveal that production? The focus here is that collider question rather than a full review of dark-matter evidence or candidate theory.

The key difficulty is that a collider dark-matter particle, if it is stable and weakly interacting on detector scales, would not be directly reconstructed by ATLAS or CMS. It would leave no ordinary charged-particle track, no normal calorimeter shower, and no clean object in the event display. The experiment would see the visible particles around it and infer the invisible component from imbalance, recoil, or unusual topology. The CMS dark-sector review states this logic in modern form: models with invisible massive particles are probed through missing transverse momentum recoiling against visible Standard Model particles, while dark-sector mediators and more complex scenarios can also produce visible final states and long-lived-particle signatures [1].

This makes collider dark-matter searches an inference program before they are a discovery narrative. A search begins by reconstructing visible objects, estimating whether the transverse momentum in the event balances, controlling Standard Model processes that mimic the signature, and asking whether the data depart from that expectation. The ATLAS missing-transverse-momentum performance work is therefore not a technical side issue. It is part of the physics argument, because missing transverse momentum is constructed from calibrated objects, soft activity, and procedures designed to control ambiguity in the event reconstruction [2]. A claim about invisible particles depends on the quality of the visible reconstruction.

The search program also cannot stop at a single missing-momentum channel. If a mediator connects dark matter to ordinary matter, that mediator may produce both invisible and visible signatures. The LHC Dark Matter Working Group's heavy-mediator recommendations explicitly compare searches with missing momentum and searches without missing momentum for the same broad mediator framework [3]. This is the point at which a simple "missing energy" story becomes a richer collider program. Monojet searches, mono-Higgs searches, visible resonances, top-associated production, semi-visible jets, emerging jets, and long-lived-particle searches are not isolated curiosities. They are different ways of making hidden physics experimentally legible.

The central claim is that LHC dark-matter searches are best understood not as direct attempts to see dark matter, but as inference strategies that convert invisible final states into constrained detector signatures. The field has moved from simple missing-energy searches toward a broader program of mediator constraints, visible-invisible complementarity, top-associated channels, dark-sector showers, semi-visible jets, and long-lived-particle signatures. That movement should not be read as confusion or trend-chasing. It follows from treating dark matter as a set of experimentally testable production and decay hypotheses rather than as one fixed detector object.

The argument proceeds through that logic in stages. It first distinguishes public shorthand about missing energy from the technical observable of missing transverse momentum. It then explains mono-X searches and simplified mediator models, before turning to visible-invisible complementarity and dark-sector signatures such as semi-visible and emerging jets. The final section explains why null results are still scientific outputs: they constrain benchmark models, reveal where signatures are already tightly bounded, and motivate broader searches in regions that simpler selections may miss.

Table 1. Signature Taxonomy For LHC Dark-Matter And Dark-Sector Searches

Search family	Main observable	Detector-level logic	What it tests	Example source
Missing transverse momentum	Event-level transverse imbalance	Reconstruct all visible objects and soft activity, then test whether the event balances in the transverse plane.	Invisible particles, neutrino backgrounds, mismeasurement control, and reconstruction quality.	[2]; [4]

Search family	Main observable	Detector-level logic	What it tests	Example source
Monojet	Hard jet plus large missing transverse momentum	A high-momentum jet recoils against invisible momentum; leptons and photons may be vetoed to control backgrounds.	Classic invisible-particle production, compressed scenarios, and broad new-physics benchmarks.	[5]
Mono-X electroweak channels	Photon, W, Z, or Higgs recoil plus missing transverse momentum	The visible boson sharpens the event category and can encode coupling assumptions.	Mediator structure and associated production beyond generic initial-state-radiation recoil.	[6]; [7]
Visible mediator searches	Dijet, dilepton, or other visible resonance signatures	A mediator that couples to Standard Model particles may decay visibly instead of invisibly.	Visible-invisible complementarity, mediator mass, width, and Standard Model couplings.	[3]
Dark-Higgs associated production	Dark-sector Higgs candidate plus missing momentum and Standard Model decay products	A richer dark-sector model produces visible decay chains rather than only a single recoil object.	Dark-sector mass scales, boosted reconstruction, and Higgs-sector portal assumptions.	[8]
Top-associated dark matter	Top-quark final states plus missing transverse momentum	Heavy-flavor objects make coupling assumptions more specific than inclusive monojet searches.	Top-philic or heavy-quark-sensitive mediator structures.	[9]
Semi-visible jets	Jet-like object partly aligned with missing transverse momentum	A hidden shower contains both visible and invisible components inside or near the same jet topology.	Strongly interacting dark sectors and dark shower fractions.	[10]
Emerging jets	Jets with multiple displaced vertices	Long-lived dark hadrons decay back into Standard Model particles after traveling measurable distances.	Dark confinement, mediator production, and lifetime-dependent signatures.	[11]
Long-lived-particle searches	Displaced, delayed, disappearing, or otherwise non-prompt signatures	Hidden-sector particles may decay after macroscopic detector travel rather than promptly or never.	Lifetime, detector timing/tracking/vertexing reach, and unconventional backgrounds.	[1]

2. From Missing Energy To Missing Transverse Momentum

The phrase "missing energy" is useful as a public doorway into collider dark-matter searches, but it is not precise enough to carry the technical argument. At the LHC, ATLAS and CMS do not begin with a direct image of a dark-matter particle leaving the detector. They begin with a balance problem. The incoming protons collide through partons whose longitudinal momenta are not known event by event, so the experimentally useful conservation test is built in the plane transverse to the beams. If the visible reconstructed particles in that transverse plane do not balance, the event may contain invisible particles, badly measured objects, detector noise, or ordinary Standard Model processes with neutrinos. For that reason, the observable at the center of the paper is missing transverse momentum, not a vague absence of energy.

This distinction matters because missing transverse momentum is not a single hit, object, or detector subsystem. It is an event-level reconstruction assembled from everything the detector believes it has measured. In the ATLAS performance study, missing transverse momentum is reconstructed from calibrated electrons, muons, photons, hadronically decaying tau leptons, jets, and soft hadronic activity, with explicit procedures to avoid double counting signals that could otherwise be assigned to more than one reconstructed object [2]. That reconstruction logic turns the detector from a collection of subsystems into a momentum-balance instrument. The claim that an event has large missing transverse momentum is therefore a statement about the coherence of the whole reconstructed event, not merely about an empty region of the detector.

CMS approaches the same problem through its own reconstruction chain. Its 13 TeV missing-transverse-momentum performance paper evaluates the scale and resolution of `pTmiss`, studies anomalous missing-momentum events, and describes mitigation of pileup effects using pileup-per-particle identification [4]. The key lesson is that missing transverse momentum must be defended against ordinary detector and collision complications before it can be interpreted as a possible new-physics signature. Multiple proton-proton interactions in the same bunch crossing can add soft activity; jet energy mismeasurement can fake imbalance; detector pathologies can produce anomalous tails; and neutrinos produce real missing momentum inside Standard Model processes. A dark-matter search that ignores these complications would not be ambitious. It would simply be under-controlled.

The detector-level nature of missing transverse momentum also explains why the LHC dark-matter program cannot be described as "looking for dark matter" in the same way that a tracking detector looks for charged-particle trajectories. If a weakly interacting particle escapes, it does not leave a track in the inner detector, does not deposit a normal electromagnetic or hadronic shower, and does not behave like a reconstructed muon. It enters the analysis through imbalance. The visible part of the event, and the reliability of its reconstruction, is what gives the invisible part any experimental meaning. This is the first major inference step in the paper's argument: collider dark-matter searches make absence measurable only by measuring everything else well.

That inference step becomes concrete in monojet-style searches. The ATLAS search for new phenomena in events with an energetic jet and large missing transverse momentum used 139 fb^-1 of 13 TeV proton-proton collision data from 2015-2018, requiring at least one high-momentum jet and vetoing reconstructed leptons and photons [5]. Several signal regions were then defined with increasing missing-transverse-momentum requirements. The paper reports overall agreement between data and Standard Model predictions, and translates that agreement into visible-cross-section limits and exclusions in multiple benchmark scenarios, including weakly interacting dark-matter candidates [5]. The structure of the result matters: the search does not claim that every high-pTmiss event is suspicious. It builds a controlled event selection, predicts backgrounds, compares data to the prediction, and then constrains models when no statistically significant excess appears.

This is why missing transverse momentum should be treated as a method, not a slogan. It is powerful precisely because it is indirect. It can register neutrinos from known processes, invisible particles from possible new processes, and detector failures that must be removed or modeled. That ambiguity is not a weakness if the analysis makes it explicit. In fact, the scientific strength of the LHC program comes from turning the ambiguity into a disciplined search architecture: define the visible objects, reconstruct the event-level imbalance, control the backgrounds, test the tails of the distribution, and interpret deviations or non-deviations within specific models. The invisible particle is never the first object in the chain. It is the final interpretation of a carefully audited imbalance.

The next step is to ask what visible object makes that imbalance searchable. A large missing-transverse-momentum event by itself is too broad to define a dark-matter program. Mono-X searches sharpen the problem by requiring a visible recoil object: a jet, photon, weak boson, Higgs boson, heavy-flavor system, or top-quark system that balances the invisible momentum. The move from missing transverse momentum to mono-X is therefore not just a catalog of channels. It is the point where a generic imbalance becomes a model-sensitive collider signature.

3. Mono-X Searches And Simplified Models

Mono-X searches are the classic bridge between invisible-particle production and visible collider evidence. The "X" is not decorative shorthand; it is the experimentally reconstructed recoil that makes the invisible final state searchable. In a monojet search, a hard jet recoils against missing transverse momentum. In monophoton or mono-Z/W searches, an electroweak object plays the same role. In mono-Higgs or heavy-flavor associated channels, the visible object carries more model information because it can point toward the structure of the mediator's couplings. The common logic is that the detector cannot see the invisible particles, but it can test whether a visible object appears with an imbalance pattern that exceeds Standard Model expectations.

The ATLAS monojet search provides the cleanest anchor for this logic. Its selection asks for an energetic jet and large missing transverse momentum, while rejecting reconstructed leptons and photons that would point toward more ordinary electroweak backgrounds [5]. This design makes sense because a jet from initial-state radiation can provide the visible recoil against invisible particles. But it also shows why monojet searches are technically unforgiving. The dominant backgrounds are not imaginary: Z boson production with invisible decays to neutrinos, W boson production where the charged lepton is not reconstructed or fails the selection, and multijet events where mismeasurement can produce apparent imbalance. The search is therefore not a hunt for spectacular single events. It is a statistical comparison between a carefully modeled Standard Model expectation and the observed tails of missing-transverse-momentum distributions.

That statistical structure is what gives mono-X searches their scientific value even when they produce no discovery. The ATLAS monojet result reports overall agreement between the observed data and Standard Model predictions, then converts that agreement into model-independent cross-section limits and model-dependent exclusions [5]. The outcome is not a blank page. It narrows the space of possible new processes that could have produced the selected signature. In a portfolio paper, this point should be stated directly: collider dark-matter searches often work by making non-observation precise. Their output is a constrained map of what certain dark-matter or dark-sector models are not allowed to do at accessible energies and couplings.

The interpretive problem is that a missing-momentum signature alone does not define a theory. A final state with a jet and missing transverse momentum can be generated by many possible mechanisms. Early missing-energy interpretations often used effective-field-theory language, in which the mediator between the Standard Model and dark matter is too heavy to be resolved and its effects are compressed into contact interactions. That framing can be useful, but it becomes limited at LHC energies, where the mediator may be produced directly or where the assumptions behind the contact approximation may fail. The simplified-model literature responded to this problem by adding explicit mediating particles and specifying their masses, couplings, widths, and decay patterns [6]. A simplified model is not a complete dark sector, but it gives searches a more honest bridge between detector signature and particle interpretation.

At this point the argument moves beyond the phrase "missing energy." If the mediator is explicit, then the LHC is not only testing invisible final states. It is also testing the particle that may connect visible matter to the dark sector. The 2014 simplified-model report emphasizes that missing-energy searches should be interpreted broadly and that additional mediator particles can appear as final states in a coherent interpretation [6]. The operational point is that adding a mediator does not merely decorate the Feynman diagram. It changes what the experiment should look for. A mediator may produce missing momentum when it decays to dark matter, but it may also generate visible resonances or other final states if it couples back to Standard Model particles.

The LHC Dark Matter Working Group's heavy-mediator recommendations sharpen this point. For vector and axial-vector mediators, the same interaction structure that produces invisible mono-X signatures can also produce visible signals without missing momentum through the same production vertices [3]. The practical consequence is that missing-momentum searches and visible-channel searches are complementary, not rival stories. A model excluded weakly in a monojet channel may be constrained strongly in a dijet or dilepton channel, depending on mediator mass, width, and couplings. Conversely, invisible decays can dominate in parts of parameter space where visible resonance searches are less sensitive. The correct unit of interpretation is the model's signature pattern across channels, not one plot in isolation.

This complementarity also prevents a common misunderstanding. It is tempting to describe mono-X results as if they directly measure dark matter's existence or absence. They do not. They test benchmark relationships among production rate, visible recoil, invisible momentum, mediator structure, and background expectations. The inference is model-dependent by construction. That does not make it weak; it makes it scientifically legible. A clean benchmark allows ATLAS and CMS to compare analyses, combine or contrast channels, and translate detector-level results into exclusion contours that theorists and other experiments can read.

The evolution toward richer simplified models shows why this machinery keeps changing. The LHC Dark Matter Working Group's spin-0 report notes that simplified models helped sharpen the LHC dark-matter program, especially in how results are presented and compared with direct- and indirect-detection experiments, but also emphasizes limitations such as restricted phenomenology and theoretical consistency concerns [7]. Its discussion of the `2HDM+a` benchmark shows how a more theoretically structured model can generate a richer mono-X pattern and a wider set of relevant constraints [7]. In other words, the search program did not abandon mono-X logic; it made the logic more conditional, more explicit, and more connected to the model assumptions underneath it.

This section sets up the modern turn in collider dark-matter searches. Once mediator structure is taken seriously, the experimental landscape naturally expands beyond a single hard jet and missing transverse momentum. The visible object can be a Higgs boson, a heavy-flavor pair, or a top-quark system. The invisible state can be part of a larger dark sector rather than a stable WIMP-like particle. The next section should therefore treat visible-invisible complementarity as the hinge between classic mono-X searches and newer dark-sector signatures. The real story is not that the LHC moved from one fashionable channel to another. It is that each added signature makes a different part of the invisible-production hypothesis testable.

4. Visible-Invisible Complementarity

Once a dark-matter search introduces an explicit mediator, the paper can no longer treat missing transverse momentum as the whole experimental story. A mediator is a particle hypothesis, not a detector signature by itself. If it couples to quarks, leptons, gauge bosons, or Higgs-sector states, it can produce different visible and invisible final states depending on its mass, width, couplings, and available decay modes. The same mediator that appears in a mono-X diagram as the bridge to invisible particles may also be constrained by visible resonances or associated production channels. That is why the LHC Dark Matter Working Group framed heavy-mediator comparisons around both invisible and visible decay channels rather than treating mono-X searches as self-contained [3].

This complementarity is more than a bookkeeping preference. It changes what a limit means. A monojet result can constrain the rate at which a mediator produces invisible particles recoiling against a jet, but it does not by itself determine whether the mediator also decays visibly, whether it couples preferentially to leptons or quarks, or whether its branching fractions make another channel more sensitive. The Working Group's vector and axial-vector mediator recommendations are important because they ask ATLAS and CMS results to be compared across signatures that share the same underlying interaction structure [3]. A missing-momentum search and a dilepton or dijet resonance search may look different at detector level, but in a mediator model they can be different projections of the same physics.

The value of this cross-channel view is clearest when the paper resists a false choice between "invisible" and "visible" searches. Invisible searches are powerful when dark-sector particles leave the detector without interacting, but visible searches can become decisive when the mediator has appreciable Standard Model decays. A parameter point might evade a missing-momentum selection because the invisible branching fraction is small, while still being constrained by a visible resonance. Another point might evade visible searches because the mediator decays dominantly into the dark sector, while becoming vulnerable in monojet or mono-Higgs channels. Complementarity therefore keeps the interpretation honest: a benchmark model is tested by its whole signature pattern.

Spin-0 mediator models show why this matters for theory as well as analysis. The LHC Dark Matter Working Group's next-generation spin-0 report credits simplified models with sharpening LHC dark-matter interpretation, but also notes limitations such as restricted phenomenology and consistency concerns [7]. Its `2HDM+a` benchmark is useful precisely because it generates a richer pattern of mono-X and non-mono-X signatures than the simplest pseudoscalar mediator picture [7]. This does not require a full two-Higgs-doublet derivation. The relevant point is that model structure determines which detector signatures deserve attention. A model with more realistic Higgs-sector structure can make Higgs-associated, heavy-flavor, or other channels central rather than peripheral.

The ATLAS dark-Higgs search gives a concrete example. Instead of searching only for an invisible system recoiling against a jet, ATLAS considered dark-matter production in association with a dark Higgs boson decaying to `W+W-`, using 139 fb^-1 of 13 TeV `pp` collision data in a one-lepton final state [8]. The analysis reconstructed hadronic `W` decays using calorimeter jets or track-assisted reclustered jets, a detail that matters because boosted decay topologies are part of the detector challenge [8]. The observed data agreed with Standard Model predictions, and the result excluded dark-Higgs masses between 140 and 390 GeV in the studied scenarios [8]. This is still a dark-matter search, but it is no longer reducible to a simple "jet plus missing momentum" picture.

Top-associated searches make a related point from a different direction. Because the top quark is heavy and closely tied to electroweak-symmetry-breaking physics, final states with top quarks can probe dark-matter coupling structures that generic monojet searches may blur. A review of dark-matter searches with top quarks describes ATLAS and CMS searches in final states with top quarks using the full LHC Run 2 dataset [9]. For this paper, top-associated production should remain a focused example rather than a side paper. Its job is to show that the visible object in a dark-matter search is not just a trigger handle; it can encode assumptions about how the dark sector talks to the Standard Model.

Visible-invisible complementarity is therefore the hinge between classic missing-momentum searches and the broader dark-sector program. Missing transverse momentum tells the experiment that something in the transverse balance is unaccounted for. Mono-X signatures add a visible recoil. Mediator models explain why different visible and invisible channels can constrain the same hypothesis. But once the dark sector is allowed to contain multiple particles, hidden showers, displaced decays, or long-lived states, the search program must become even wider. The next step is to examine signatures where the problem is not simply an invisible recoil, but an entire hidden event structure partially returning to the detector.

5. Beyond Prompt Missing Energy: Dark Sectors And Long-Lived Particles

The phrase "dark sector" marks a real expansion of the search problem. In a simple missing-momentum benchmark, the dark matter candidate is stable on detector scales and escapes, leaving visible recoil as the main handle. In a dark-sector scenario, the hidden side may contain multiple particles, new forces, mediator states, showering behavior, and lifetimes long enough to change where and how the detector sees the event. The CMS dark-sector report captures this broadening by reviewing searches not only for invisible massive particles and mediators, but also for more complex dark sectors with multiple new particles and long-lived particles that can generate unconventional signatures [1]. The experimental question is no longer only "what recoils against the invisible object?" It becomes "what detector topology would a hidden sector leave if some of it escapes, some of it decays back, and some of it does so late?"

This shift matters because it keeps collider dark-matter searches from being trapped inside a single WIMP-like picture. Missing transverse momentum remains central, but it is no longer the only organizing observable. A dark-sector event may have ordinary-looking jets mixed with invisible components, displaced vertices inside jets, delayed decays, anomalous shower shapes, or final states that only become distinctive when reconstruction algorithms look for patterns beyond prompt objects. The scientific logic remains the same as in the earlier sections: invisible physics must be inferred from visible consequences. What changes is the richness of those consequences.

Semi-visible jets are one of the clearest examples because they sit directly between ordinary hadronic activity and missing momentum. ATLAS describes semi-visible jets as arising in strongly interacting dark sectors, where a significant contribution to the event's missing transverse momentum can be associated with a jet-like object [10]. In the Run 2 non-resonant semi-visible-jet search, the topology can place one jet in alignment with the missing-transverse-momentum direction, a pattern that would be strange for the simplest monojet intuition but natural for a shower containing both visible and invisible particles [10]. That alignment detail is exactly the sort of detector-level signature this paper is trying to foreground.

The ATLAS semi-visible-jet result also illustrates how dark-sector searches remain disciplined even when the signatures sound exotic. The analysis used 139 fb^-1 of 13 TeV proton-proton collision data from Run 2 and presented the first search for semi-visible jets produced through a `t`-channel mediator exchange [10]. No excess over Standard Model predictions was observed. Under the stated benchmark assumptions, mediator masses up to 2.7 TeV were excluded at 95% confidence level for coupling strength equal to one, and upper limits on the coupling were also derived [10]. The key point is not the numerical limit by itself. The key point is that a complicated hidden-shower idea was turned into a well-defined topology, event selection, background comparison, and exclusion statement.

Emerging jets push the same principle into the lifetime dimension. In the 2025 ATLAS emerging-jet search, the hypothetical dark sector contains dark quarks charged under a confining gauge group and coupled to the Standard Model through a new mediator [11]. These dark quarks shower and hadronize into long-lived dark mesons that later decay back into Standard Model particles, producing jets with multiple displaced vertices [11]. The phrase "emerging jet" is therefore not just a dramatic label. It names a reconstruction problem: the visible activity appears with spatial structure inconsistent with an ordinary prompt jet.

The emerging-jet analysis used 51.8 fb^-1 of 13.6 TeV proton-proton collision data collected by ATLAS during 2022 and 2023, targeting events with pairs of emerging jets produced through either an `s`-channel vector mediator or a `t`-channel scalar mediator [11]. Again, no significant excess over the Standard Model background was observed. The result excluded ranges of vector-mediator and scalar-mediator masses for specified coupling choices and dark-pion proper decay lengths [11]. For this paper's argument, the important feature is that lifetime became part of the search space. Instead of assuming every relevant particle is prompt or invisible, ATLAS built a search around displaced structure inside jets.

Long-lived-particle signatures matter for dark sectors because they loosen the assumption that new particles must either decay immediately or escape entirely. A particle with a centimeter-scale or meter-scale lifetime may leave a displaced vertex, a delayed object, a disappearing or emerging signature, or a pattern that standard prompt-object selections were not designed to capture. The CMS dark-sector review emphasizes that many alternative scenarios include long-lived particles and that such particles can create striking signatures with relatively small Standard Model backgrounds [1]. This does not make discovery easy. It makes the search program broader and more specialized, because detector timing, tracking, vertexing, and trigger strategies become part of the dark-matter question.

The dark-sector turn therefore changes the reader's mental model of collider dark matter. The early picture is a clean recoil: visible object on one side, invisible momentum on the other. The modern picture is a family of inference problems. Some signatures remain prompt and missing-momentum dominated. Others combine visible and invisible shower components. Others rely on displacement, lifetime, or anomalous hadronic structure. A professional paper should not treat these as a grab bag. They belong together because each asks how hidden physics can be made experimentally legible through detector-level structure.

That broader signature space also changes how null results should be interpreted. If ATLAS or CMS does not see an excess in a monojet search, that does not eliminate dark-sector models with semi-visible showers or displaced decays. If an emerging-jet search excludes one mediator-lifetime benchmark, that does not exhaust all hidden-valley or dark-QCD scenarios. Each result constrains a defined patch of model space. The scientific value of the LHC program is cumulative: different searches cover different ways that an invisible or hidden sector might leak information back into the detector. The next section should therefore explain how limits, exclusions, and "no significant excess" statements become useful scientific outputs rather than anticlimactic endings.

6. What Null Results Mean

The most common sentence in a collider dark-matter paper is also the easiest to misread: no significant excess over the Standard Model prediction is observed. In popular terms, that can sound like the experiment found nothing. In the logic of an LHC search, it is the beginning of the result, not the end. Once an analysis defines a signal region, estimates backgrounds, quantifies systematic uncertainties, and compares the data to the prediction, agreement becomes a constraint. It says that any new process producing that signature must be small enough, rare enough, heavy enough, weakly coupled enough, short-lived enough, long-lived enough, or otherwise different enough to evade the tested selection. A null result is therefore not empty. It is a boundary drawn around a benchmark hypothesis.

The ATLAS monojet search shows the structure clearly. After selecting events with an energetic jet and large missing transverse momentum, ATLAS found overall agreement between data and Standard Model predictions, then translated that agreement into model-independent visible-cross-section limits and exclusions in several benchmark scenarios [5]. This pattern is common because the experiment is not only asking whether one spectacular event appears. It is asking how many events of a defined type should appear under known physics, how uncertain that expectation is, and whether the observed distribution is compatible with it. If the data follow the background model, the analysis can still rule out new-physics rates that would have produced a visible excess.

The same logic applies to more structured dark-sector searches. The ATLAS dark-Higgs result found agreement with Standard Model predictions and excluded dark-Higgs masses between 140 and 390 GeV in the studied benchmark scenarios [8]. The semi-visible-jet search reported no excess and excluded mediator masses up to 2.7 TeV for a specific coupling benchmark [10]. The emerging-jet search found no significant excess and excluded ranges of vector- and scalar-mediator masses for specified couplings and dark-pion decay lengths [11]. These are not interchangeable limits. Each one lives inside the assumptions of its model, final state, reconstruction strategy, and statistical procedure.

That model dependence is not a flaw to hide. It is a condition of honesty. A monojet exclusion does not automatically eliminate a semi-visible-jet model. A semi-visible-jet limit does not cover all long-lived dark-hadron possibilities. An emerging-jet constraint is strongest for the lifetime, mass, and coupling patterns it was designed to test. A professional interpretation should therefore avoid two opposite mistakes. The first mistake is to dismiss null results as failures. The second is to overstate them as global statements about dark matter. Collider searches constrain defined signatures under defined assumptions. Their strength comes from making those assumptions explicit enough that different results can be compared, combined, or used to motivate the next search.

This is also why the shift from missing energy to dark sectors should not be framed as a replacement story. The field did not simply abandon monojet searches because they failed to discover dark matter. Rather, the accumulation of strong prompt missing-momentum constraints made it increasingly important to test signatures that simple recoil searches could miss. The CMS dark-sector report reflects this broader program by placing invisible-particle searches, visible mediator searches, complex dark-sector scenarios, and long-lived-particle signatures in a shared review frame [1]. The search space expands because the hidden sector might not look like the simplest benchmark, not because the earlier benchmarks were meaningless.

Null results also have a practical design consequence. They teach experiments where backgrounds are well controlled, where trigger thresholds matter, where reconstruction methods need refinement, and where future analyses should look for less standard topologies. A missing-transverse-momentum result can motivate visible mediator searches. A visible resonance limit can force attention back to invisible branching fractions. A semi-visible-jet or emerging-jet search can show how tracking, vertexing, machine learning, or anomaly-aware selections might open regions of hidden-sector phenomenology that prompt-object analyses do not cover. In that sense, the LHC dark-matter program is cumulative even without a discovery claim.

The paper's central argument can now be stated in operational form. ATLAS and CMS do not make dark matter visible by seeing it directly. They make invisible or hidden physics testable by building chains of inference: reconstructed objects define missing transverse momentum; visible recoil objects define mono-X signatures; mediator models connect invisible and visible channels; dark-sector searches add showering, displacement, lifetime, and topology; statistical comparisons turn agreement or disagreement into limits. If a discovery comes, it will enter through that chain. If no discovery comes, the same chain still maps what particular forms of invisible matter are allowed to be.

7. Conclusion

The LHC does not search for dark matter by waiting for an invisible particle to announce itself. It searches by constructing situations in which invisible or hidden physics would disturb the visible event in a measurable way. Missing transverse momentum turns imbalance into an observable. Mono-X searches turn invisible production into a recoil problem. Mediator models connect missing-momentum channels to visible searches. Dark-sector searches extend the same logic to semi-visible showers, displaced structures, emerging jets, and long-lived particles.

The resulting picture is sharper than the phrase "missing energy" suggests. Collider dark-matter physics is not a single channel or a single candidate model. It is a layered inference system in which reconstruction, backgrounds, triggers, object definitions, model assumptions, and statistical limits all matter. A clean null result can be as informative as a dramatic event display because it tells the field which benchmark signatures are already constrained and where future searches must become more creative.

This is why the movement from missing energy to dark sectors should be seen as methodological growth. As ATLAS and CMS constrain simple prompt signatures, the search expands toward mediator complementarity, hidden showers, lifetimes, and unusual detector topologies. Invisible matter becomes experimentally tractable only when absence is made quantitative, displacement is reconstructed, topology is classified, and model assumptions are stated clearly. The search for dark matter at the LHC is therefore not only a search for a particle. It is a demonstration of how modern collider experiments turn what they cannot see directly into physics they can test.

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